AIMO Proof Pilot — Final Grade

Problem (#_ID): 3_SINDEX **Submission:** model_7

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Final mark

Final mark (choose one): $\boxed{6}$

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

An essentially complete solution matching the official approach. It establishes the polynomial criterion, uses coprime cyclotomic factors and coefficient signs to rule out index ≥ 6, gives explicit valid partitions attaining index 5 (so $N = 5$), proves $M = 16$, and eliminates the remaining coefficient triples to reach the minimal size 32. The genuine minor error is that passing from $c_3 > 0$ to $c_3 \geq 1$ requires the integer coefficients of $h = f/L_5$, which is used but not justified (`ME2`). (CG additionally flags an unused and false general claim that a nonnegative h makes every quotient of L_5h nonnegative, around lines 148–151, an `ME3`-type slip; since it is never relied on, it does not affect the argument.) Essentially complete with a minor error, so scores **6**.